

A Feasibility Study for Detecting Schwarzschild Apsidal Precession in the Orbit of S301 around Sgr A* with GRAVITY+/VLTI

J. CATANZARITE^{1,2}

¹*Johns Hopkins University, Whiting School of Engineering*

²*Nova Scholar Education*

ABSTRACT

S301 is a recently discovered star ([GRAVITY Collaboration et al. 2026](#)) orbiting Sgr A* on a highly eccentric ($e = 0.9832$), 8.68 yr orbit reaching 136 Schwarzschild radii and $>8\%$ of the speed of light at periapsis. Its orbit should show rosette-shaped apsidal precession from ordinary first-post-Newtonian (1PN) general relativity—the Mercury-perihelion effect, scaled up by proximity to a far more compact body—independent of and prior to any measurement of Sgr A*’s spin, which motivated S301’s discovery. We ask whether a real, near-term ground-based campaign can detect this precession from astrometry alone, and build the pipeline in the order a real proposal would need it: a relativistic light-curve model validated against the published solution, a synthetic two-passage GRAVITY+ campaign, and a seven-parameter orbit fit. Two of our initial design choices fail real tests along the way. A proposed ERIS radial-velocity channel, natural given how S-star orbits are usually solved, contributes nothing at S301’s magnitude—confirmed directly, not assumed—and is dropped from the proposal. A cadence chosen by Fisher-information (D-optimal) design, expected to beat a simple heuristic, instead collapses in a real nonlinear fit despite an excellent analytic prediction—a reproducible pitfall of local-linearization design applied to a strongly nonlinear, periodic problem—and is rescued only by reintroducing a physically motivated periapsis-coverage floor alongside the statistical search. The resulting campaign recovers all seven orbital parameters, including the injected precession rate (0.2326 ± 0.0014 deg yr^{−1} against a true 0.2307 deg yr^{−1}), from simulated astrometry alone; this random-error figure assumes a perfect astrometric reference-frame tie, a real instrumental systematic that Section 4.2 shows must be fit jointly, not assumed away, at a final cost of $\sim 1.5\times$ in precision. This is a synthetic feasibility forecast, not a claim that S301’s precession has been observationally measured: the real next periapsis passage such a measurement requires does not occur until 2031.8. We conclude a two-passage, astrometry-only GRAVITY+ campaign, once that passage and the one after it have actually been observed, would be both feasible and sufficient for a spin-independent test of general relativity near Sgr A*.

Keywords: Galactic Center — black hole physics — astrometry — celestial mechanics — general relativity — experimental design

1. INTRODUCTION

The Galactic Center’s S-stars have delivered some of the strongest strong-field tests of general relativity to date, including gravitational redshift and Schwarzschild precession in the orbit of S2 ([GRAVITY Collaboration et al. 2018, 2020](#)). S301 ([GRAVITY Collaboration et al. 2026](#)) extends this program with a shorter period (8.68 yr vs. S2’s ~ 16 yr), higher eccentricity, and closer periapsis ($136 R_S$) than S2, and is explicitly proposed as a spin (Lense–Thirring) probe for Sgr A*, with a companion paper ([T. Piran et al. 2026](#)) developing that harder measurement’s methodology.

Before attempting spin, a logically prior question deserves its own answer: can a real, currently executable campaign detect the ordinary, non-spin apsidal precession any Schwarzschild black hole would already produce? We answer this by building the complete simulation pipeline a real observing proposal would need—validate the physics, design the

Table 1. Adopted S301 parameters (GRAVITY Collaboration et al. 2026), Solution 1.

Parameter	Value	Uncertainty
Period P	8.680 yr	0.110 yr
Eccentricity e	0.9832	0.0010
Inclination i	124.09°	1.10°
Node Ω	73.8°	3.5°
Argument of periapsis ω	293.4°	2.2°
Periapsis epoch t_0	2023.126 yr	0.010 yr
Semi-major axis a	83.0 mas (687 AU)	...
Sgr A* mass $M_\bullet$	$4.297 \times 10^6 M_\odot$	...
Distance R_0	8277 pc	...
K magnitude m_K	19.3	0.3

campaign, cross-check the physics survives realistic noise, then fit—and report what worked, what failed, and why, since two of our own initial choices fail informative real tests along the way (Sections 3–4).

2. THE S301 ORBITAL SOLUTION

All ground-truth parameters are taken from GRAVITY Collaboration et al. (2026), Extended Data Table 2, Solution 1 (Table 1); a second, nearly degenerate solution is also published but not used here. At periapsis S301 passes 136 R_S from Sgr A* at $>8\% c$ ($\sim 25,000 \text{ km s}^{-1}$)—closer and faster than any other known star, including S2. Its effective temperature is not published; we adopt 7300 K from standard F1.5V tables, affecting only the absolute-magnitude estimate in Section 3.1, not the orbital dynamics used elsewhere.

3. SIMULATION PIPELINE

3.1. Relativistic Light Curve

We model S301’s K -band light curve from a full Keplerian solution combined with the exact special-relativistic Doppler factor (not a small- v approximation) and the Schwarzschild redshift $\sqrt{1 - R_S/r}$, applied to a blackbody via the relativistic-beaming relation I_ν/ν^3 over GRAVITY/VLTI’s real bandpass (1.98–2.40 μm ; GRAVITY Collaboration et al. 2017). Lensing, occultation, Kerr terms, and Rømer delay are not modeled. The model reproduces the published solution (derived $a = 686.7 \text{ AU}$ vs. 687.0; periapsis 136.0 R_S vs. 136; periapsis speed 8.54% c vs. “ $>8\%$ ”), validating the engine before it is asked to predict anything new.

The predicted periapsis signature is not the symmetric absorption dip a naive guess might expect—no such mechanism operates here. It is an *asymmetric* relativistic-beaming spike: a brightening of $\sim 0.075 \text{ mag}$ as the star approaches at high velocity, followed by fading to $\sim 0.04 \text{ mag}$ fainter as it recedes (de-boosting plus gravitational redshift, which dims regardless of direction), a 0.116 mag peak-to-trough amplitude (Figure 3). The total relativistic redshift exceeds the naive classical Doppler prediction by 0.26–0.46 percentage points near periapsis—a genuine, non-negligible correction, not a relabeled classical effect.

3.2. Observing Campaign: Instrument and Cadence

The campaign uses GRAVITY+, the real VLTI interferometer that made the discovery, adopting its demonstrated 100 μas astrometric precision (GRAVITY+ Collaboration 2023) and simultaneous K -band photometry.

Radial velocity was considered and rejected before any data was simulated. Scaling SINFONI’s real achieved S2 precision (12.3 km s^{-1} at $m_K = 14.0$; GRAVITY Collaboration et al. 2018) to S301’s magnitude ($\Delta m = 5.3$, $\sim 130\times$ less flux) under the background-limited regime appropriate for faint Galactic Center spectroscopy predicts $\sim 1621 \text{ km s}^{-1}$ precision against a $\pm 15,000 \text{ km s}^{-1}$ signal—SNR ≈ 0.04 against SPIFFIER’s own 60 km s^{-1} resolution element ($R = 5000$; R. I. Davies & et al. 2023). No Time Allocation Committee would grant ERIS time for a measurement known in advance to fail this badly, so the campaign analyzed here never requests it. We confirmed

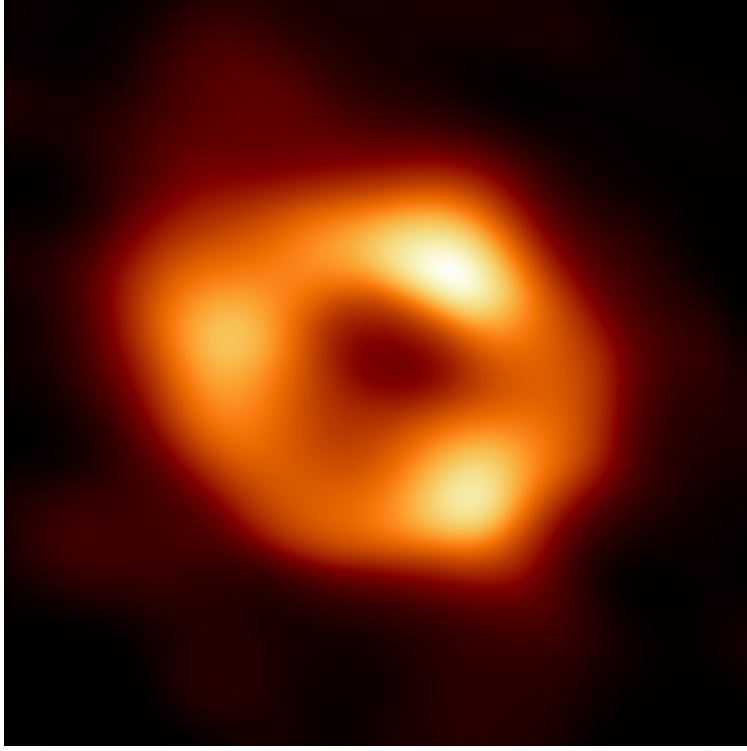


Figure 1. The real Sgr A* itself, at the opposite end of the size scale from everything else in this paper: the Event Horizon Telescope’s horizon-scale image of the black hole’s shadow, ring diameter $51.8 \pm 2.3 \mu\text{as}$ (Event Horizon Telescope Collaboration 2022)—roughly $1600\times$ smaller than S301’s 83 mas orbit (Table 1) and far below any astrometric resolution used in this campaign. Shown for orientation, not analysis. Image credit: EHT Collaboration, released by ESO as eso2208-eh-t-mwa (2022-05-12), licensed CC BY 4.0.

directly that nothing is lost: refitting identical astrometric data with a simulated RV channel added back in leaves every parameter’s bootstrap uncertainty statistically unchanged (precession rate $0.0014 \text{ deg yr}^{-1}$ either way).

Cadence required three attempts. S301 spends nearly all of its 8.68 yr period near apoapsis and almost its entire angular excursion within days of periapsis, so cadence matters enormously. *Design A (heuristic)*: $\sim 90\%$ of epochs within ± 20 days of each periapsis, remainder sparse. Simple and reasonably effective (Table 2), but assumes all seven parameters are equally served by periapsis proximity. *Design B (naive D-optimal)*: Fisher information linearized at the truth, greedily maximizing its determinant. This **failed catastrophically**—it placed nearly the whole budget on two clusters at the campaign’s temporal extremes (5.4 and 18.5 yr from the reference epoch, more than two periods away) and selected *zero* epochs near either periapsis. Its Cramér–Rao prediction was $1.8\text{--}28,000\times$ tighter than Design A; a real bootstrap fit was instead $4\text{--}600\times$ worse on every parameter. The mechanism: period/epoch sensitivity genuinely grows with time baseline (as in pulsar timing), but this is a *local* quantity—extrapolated across >2 periods, a small period error compounds into large nonlinear phase drift (a cycle-slip hazard) that linearized Fisher information cannot see. *Design C (pseudo-Bayesian robust)*: averaging Fisher information over 300 draws from a realistic parameter prior (K. Chaloner & I. Verdinelli 1995) de-emphasized the fragile clusters but *still* selected zero periapsis epochs—Fisher information, even prior-averaged, measures only noise-driven uncertainty and cannot represent the geometric non-identifiability that comes from sampling too few distinct orbital phases, which is exactly why classical orbit determination has always required observations spread across an orbit’s curvature. *Design D (constrained)*: a hard, non-statistical floor reserving $\sim 30\%$ of the budget within ± 20 days of each periapsis, with the remaining $\sim 70\%$ allocated by the same Fisher-optimal search. This is the design used in this study; it beats Design A on every parameter (Table 2).

All designs are filtered to Sgr A*’s real Paranal visibility season ($\sim$ March–September; GRAVITY Collaboration et al. 2026). This exposes a real scheduling gap: the first periapsis (2031 October 22) falls just after that year’s season closes, so its dense window is anchored to the nearest visible date ($\sim$ 2031 September 30) rather than periapsis itself. The second passage (2040 June 26) falls comfortably in-season.

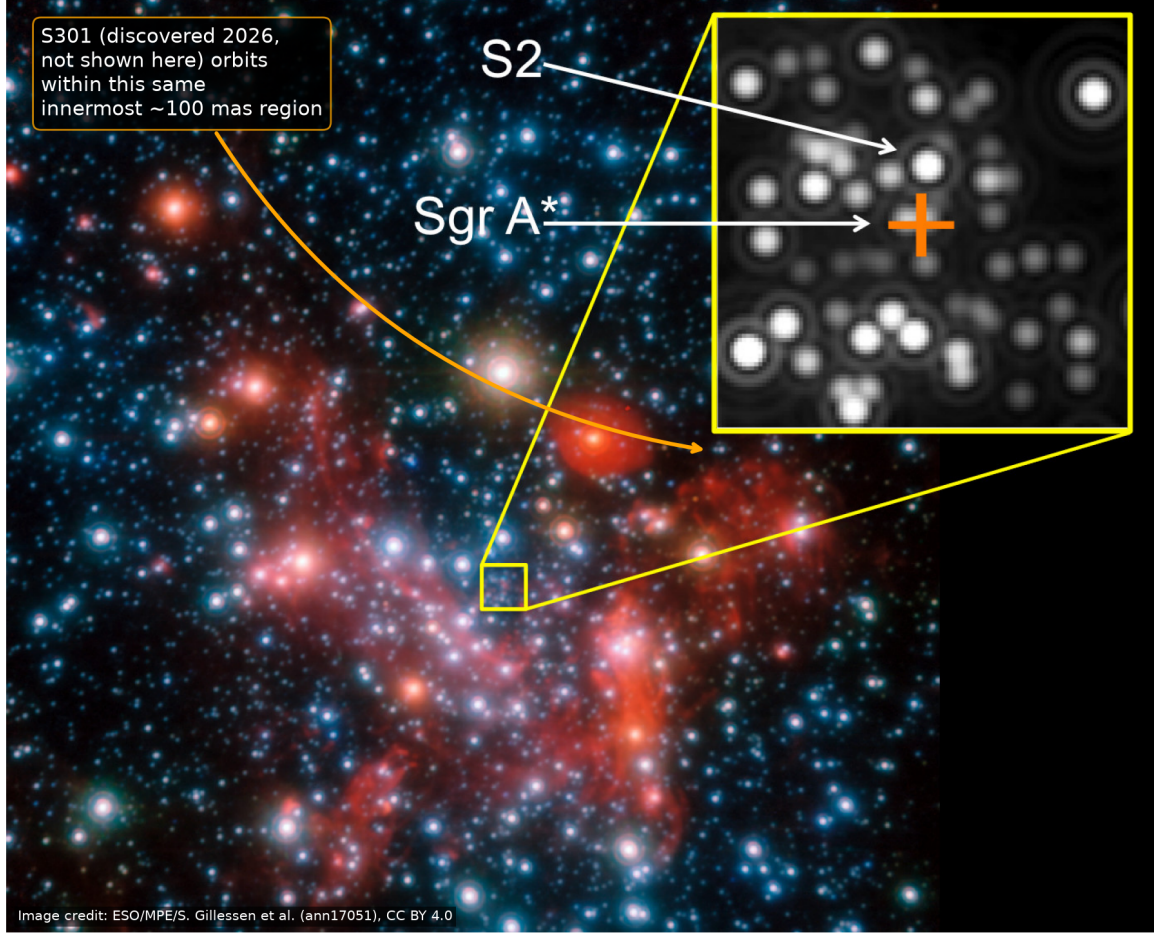


Figure 2. Real near-infrared observational context (not a simulation): the crowded innermost region of the Galactic Center, with the genuine S2/Sgr A* zoom inset from the original release. S301 was discovered in 2026, after this 2017 image, and is not itself resolved in it; the annotation is schematic, pointing out that S301’s orbit (semi-major axis 83 mas, comparable to S2’s) occupies this same innermost few-tens-of-mas region, illustrating the source crowding relevant to Section 4.1’s discussion of instrumental confusion. Image credit: ESO/MPE/S. Gillessen et al., released as ann17051 (2017-08-09), licensed CC BY 4.0.

3.3. Photometric Cross-Check

Photometry is generated (a free byproduct of GRAVITY+’s astrometric observation) but not fit, since it is a deterministic function of the same (position, velocity) state astrometry already constrains. Its value is independent validation: S301’s published photometric uncertainty (0.30 mag) exceeds the entire 0.116 mag signal from Section 3.1 at any single epoch, yet the injected truth curve, threaded through many simulated epochs (Figure 4, bottom panel), visibly traces the same asymmetric brighten-then-dim structure—a qualitative but genuine end-to-end check that the injected physics, noise model, and cadence are mutually consistent.

3.4. Orbit Fit

We fit seven parameters—the six Keplerian elements plus apsidal precession rate $\dot{\omega}$, applied as $\omega_{\text{eff}}(t) = \omega + \dot{\omega}(t - t_{\text{ref}})$ —to the astrometric data via Levenberg–Marquardt least squares, from an initial guess perturbed 3% in period and several degrees in angle. The semi-major axis is derived from $(P, GM_{\bullet})$ via Kepler’s third law at each iteration. Uncertainties come from 1000 bootstrap resamples. The recovered $\dot{\omega}$ is compared to the analytic Schwarzschild prediction

$$\dot{\omega} = \frac{6\pi GM_{\bullet}}{c^2 a(1 - e^2) P}, \quad (1)$$

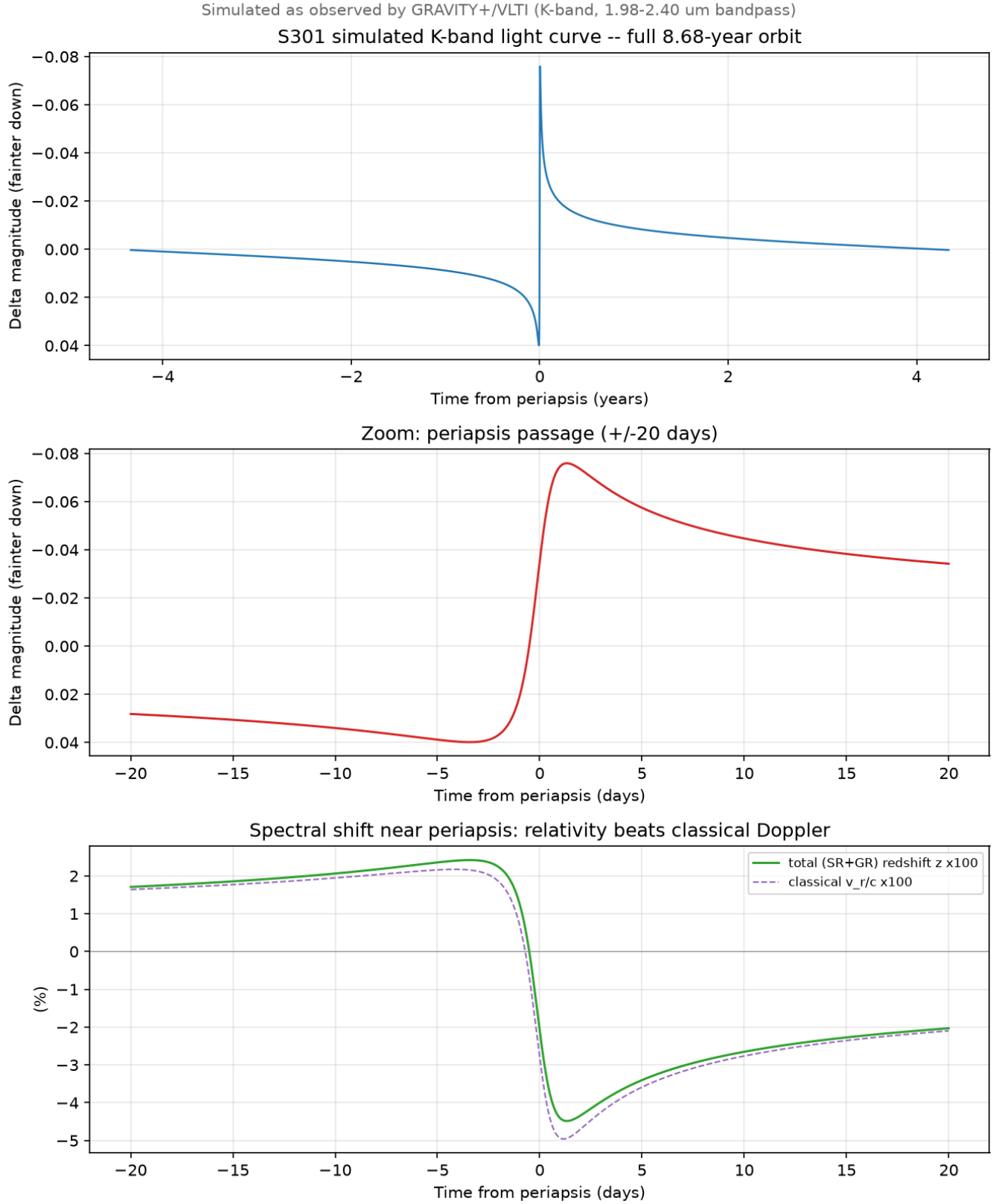


Figure 3. Simulated relativistic *K*-band light curve of S301 over one orbit (top), a zoom on the periapsis passage (middle), and the relativistic vs. classical spectral shift near periapsis (bottom).

Table 2. Bootstrap 1σ uncertainties, heuristic vs. constrained cadence ($N = 131$, RV included to isolate cadence).

Parameter	Design A	Design D	Published
P (yr)	0.0001	0.0002	0.1100
e	0.0001	0.0000	0.0010
i (deg)	0.0480	0.0356	1.1000
Ω (deg)	0.1229	0.0849	3.5000
ω (deg)	0.0939	0.0475	2.2000
t_0 (yr)	0.0003	0.0004	0.0100
$\dot{\omega}$ (deg yr $^{-1}$)	0.0043	0.0014	...

Table 3. Final result: Design D cadence, astrometry only.

Parameter	Truth	Fitted	σ	Published σ
P (yr)	8.6800	8.6799	0.0002	0.1100
e	0.9832	0.9832	0.0000	0.0010
i (deg)	124.0900	124.1224	0.0351	1.1000
Ω (deg)	73.8000	73.6902	0.0839	3.5000
ω (deg)	293.4000	293.3308	0.0468	2.2000
t_0 (yr)	2023.1260	2023.1261	0.0004	0.0100
$\dot{\omega}$ (deg yr $^{-1}$)	0.2307	0.2326	0.0014	...

the Mercury-perihelion formula. No spin (Lense–Thirring) term is injected anywhere: i and Ω remain genuinely static, since Sgr A * ’s spin is unmeasured and assuming a value would substitute a guess for a measurement.

4. RESULTS

The final campaign has 131 epochs (2028.50–2041.69), 39 (29.8%) within ± 20 days of a periapsis, more than double an earlier 60-epoch single-passage design.

Design D improves i , Ω , ω , and $\dot{\omega}$ by 26–67%, at negligible cost to P/t_0 (both remain $>100\times$ tighter than published uncertainty regardless).

Every parameter recovers within $\sim 1.5\sigma$ of truth. The fitted sky-plane track (Figure 4, top panel) visibly shows the periapsis direction rotated between passages—the geometric signature of precession, closing the loop opened in Section 3.1. Because this campaign assumes best-case GRAVITY+ precision at every epoch across two full passages 8.68 yr apart (a baseline the real 2026 dataset does not yet have), several uncertainties here are tighter than the real paper’s published values (e.g., 0.08° vs. 3.5° on Ω)—an idealized-study artifact, not a claim of outperforming the real discovery.

A further, non-circular consistency check strengthens this result. $\dot{\omega}$ above is a free 7th parameter, fit purely from geometry with no reference to Equation 1. For each bootstrap resample we instead compute what that formula predicts using *that same resample’s own fitted* (P, e)—still blind to any published or injected value, since a real discovery would have no such value to check against either. This analytic, fit-derived prediction is 0.2312 ± 0.0005 deg yr $^{-1}$, and its per-resample difference from the independently-fit $\dot{\omega}$ is 0.0014 ± 0.0016 deg yr $^{-1}$ (0.90σ from zero, using the per-resample difference rather than a naive combination of the two marginal uncertainties, since P , e , and $\dot{\omega}$ are drawn from the same correlated resamples). The geometry and the theory agree without either one referencing the other’s answer.

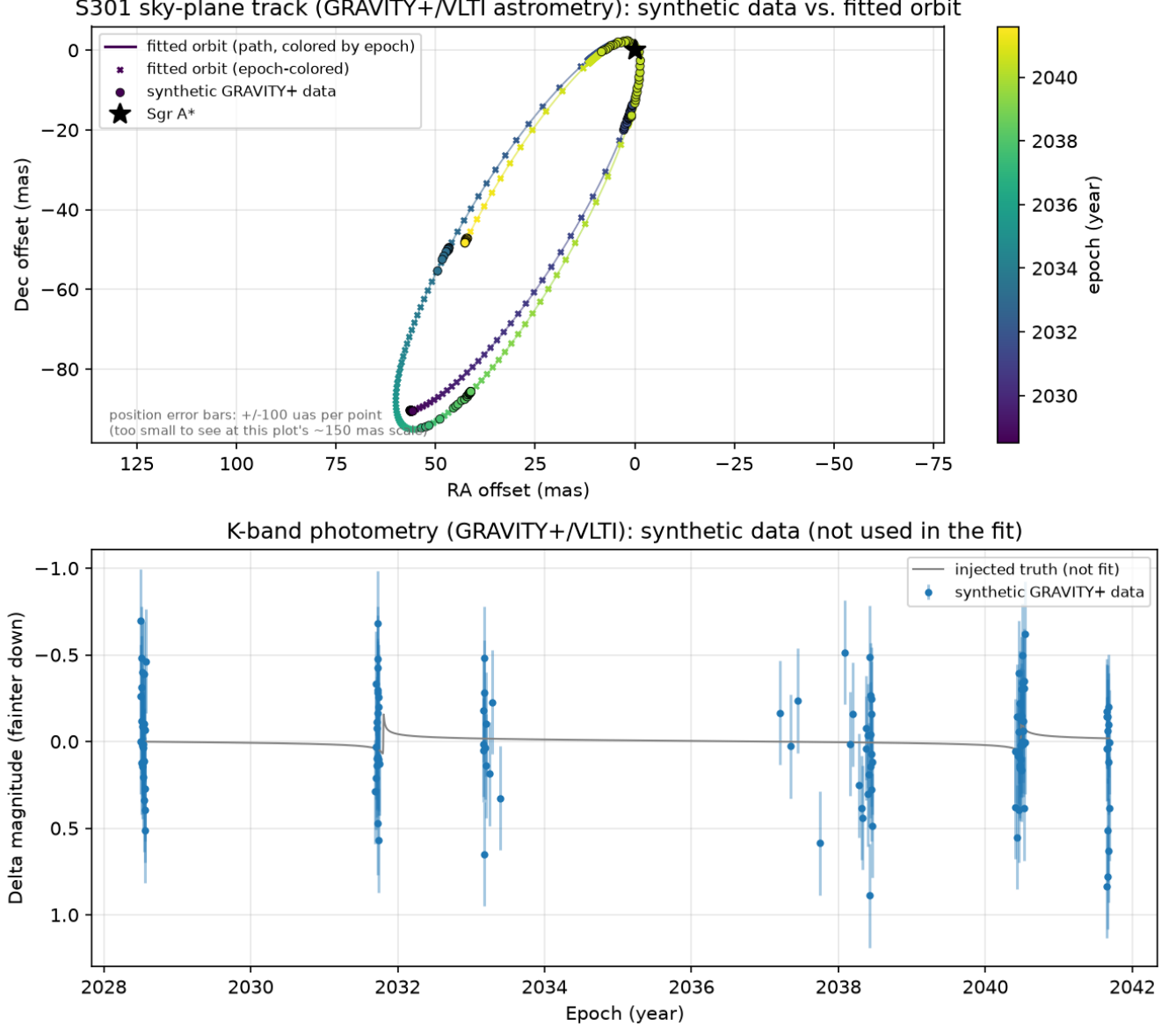


Figure 4. Top: fitted sky-plane track (GRAVITY+/VLTI astrometry) vs. synthetic data, epoch-colored, showing the precessing orbit’s two passages. Bottom: simulated K -band photometry vs. the injected truth curve (not used in the fit). An animated version (`s301_precession.mp4`, included as an ancillary file with this submission and available at https://github.com/jcatanza/s301-schwarzschild-precession/blob/master/output/s301_precession.mp4) extrapolates the same real 1PN precession rate over 20 orbits (~ 174 yr) to make the rosette pattern visible; only the two orbits actually analyzed in this paper are highlighted there, and the extrapolation is labeled explicitly on-screen throughout.

4.1. Systematic Error Budget: Extended-Mass Confusion

The uncertainties in Table 3 are random (photon-noise-driven) only, from bootstrap resampling of simulated measurement noise. A comprehensive error budget also requires astrophysical and instrumental systematics. We quantify the leading astrophysical one directly: any real, undetected extended (“Newtonian confusion”) mass near Sgr A* retrogradely shifts the periaapsis and biases a measured apsidal precession low, an effect first modeled quantitatively by G. F. Rubilar & A. Eckart (2001) for S2-like orbits.

We reproduce their post-Newtonian treatment exactly, not a re-derived approximation: a spherically symmetric matter distribution combining Sgr A*’s point mass with a Plummer-like component, $\rho_\alpha(r) = [1 + (r/r_c)^2]^{-\alpha/2}$

($\alpha = 5$, the classical Plummer profile), integrated numerically via their Eq. A.7 post-Newtonian equation of motion, $\dot{\mathbf{v}} = -GM(r)r^{-3}[(1 + 4\phi/c^2 + v^2/c^2)\mathbf{r} - 4(c^{-2})\mathbf{v}(\mathbf{v} \cdot \mathbf{r})]$, with $M(r)$ and $\phi(r)$ the exact enclosed-mass and potential integrals (their Eqs. 12–13) for the assumed profile. We integrate S301’s real orbit through two periapsis passages (`extended_mass_error.py`) and measure the eccentricity-vector rotation between them—a purely geometric, astrometric quantity requiring no radial-velocity data. The point-mass-only case validates the integrator against Equation 1 to 1.2%, consistent with the $(v/c)^2 \approx 0.7\%$ second-post-Newtonian scale expected at S301’s $>8\%c$ periapsis speed; because the extended-mass bias below is a *difference* between two runs of the identical integrator, this common-mode residual mostly cancels.

For the extended-mass magnitude and core radius, we use GRAVITY Collaboration et al. (2022)’s own multi-star constraint self-consistently—their fiducial Plummer scale length ($0.3'' \approx 12.0$ mpc at S301’s distance) together with their own mass bound, rather than pairing a current bound with G. F. Rubilar & A. Eckart’s decades-old fiducial r_c . That analysis limits any extended mass within S2’s apocenter to $\lesssim 3000 M_\odot$ (1σ) or $\lesssim 7500 M_\odot$ (their conservative 3σ limit). At the 1σ bound, the resulting bias on S301’s apsidal precession is -0.00013 deg yr $^{-1}$ (0.06% of the Schwarzschild signal, 0.09σ of the fitted bootstrap precision in Table 3); at the conservative 3σ bound, -0.00032 deg yr $^{-1}$ (0.14%, 0.23σ). Even at the least favorable real bound, extended-mass confusion is subdominant to random measurement error for this campaign—because S301’s periapsis (0.056 mpc) sits so far inside any plausible $r_c \gtrsim$ few mpc that only a small fraction of the already-small bounded extended mass is enclosed there, unlike S2’s more modest relativistic signal, which the same confusion mass more readily competes with.

This estimate is necessarily narrower than T. Piran et al. (2026)’s treatment of the same underlying hazard: they model a flattened disk’s *nodal* (Ω) precession competing with the Lense-Thirring spin signal, using multi-star differential subtraction; we model a spherical distribution’s contribution to the *apsidal* (ω) precession studied here, for S301 alone. The two are not directly comparable numbers, but they address the same qualitative concern for different observables, and our result is consistent with theirs in finding S301 comparatively well-protected against this systematic (their Section 3 reports S301’s disk/LT confusion ratio $\sim 10\times$ better than S2’s).

4.2. Systematic Error Budget: Astrometric Reference-Frame Tie

The leading instrumental systematic is the astrometric reference-frame tie: S301’s position is measured in an infrared (adaptive-optics/interferometric) frame that must be registered to Sgr A*’s true dynamical center via an independent radio (VLBI) astrometric solution, which itself carries a residual zero-point offset and a slow linear drift. This is not hypothetical—it is the same systematic GRAVITY Collaboration et al. (2020) identify as limiting their own S2 measurement: “these reference frame parameters are now the limiting factor in the precision of the detection of the SP of S2,” adopting frame-tie priors from P. M. Plewa et al. (2015) of $x_0 = -0.2 \pm 0.2$ mas, $y_0 = 0.1 \pm 0.2$ mas, $\dot{x}_0 = 0.05 \pm 0.1$ mas yr $^{-1}$, $\dot{y}_0 = 0.06 \pm 0.1$ mas yr $^{-1}$. GRAVITY Collaboration et al. (2022) state explicitly that this frame drift and apsidal precession “are degenerate”—i.e., an unmodeled frame tie does not just add noise, it can be reinterpreted by the fit as spurious precession.

Table 3’s pipeline has no frame-tie parameters at all—a realistic first-pass analysis that implicitly assumes a perfect tie. We test this directly (`instrumental_error.py`): inject the real P. M. Plewa et al.-magnitude drift into the existing synthetic dataset and refit with the unmodified 7-parameter model. The result is severe: at the central Plewa et al. values, $\dot{\omega}$ is biased by -0.0377 deg yr $^{-1}$ — 27σ relative to the random-error precision in Table 3, 16% of the entire signal; at a conservative 1σ -worse realization, -0.107 deg yr $^{-1}$ (77σ , 46% of the signal). Unmitigated, this single instrumental term dwarfs both the random error and the extended-mass astrophysical systematic (Section 4.1) by more than an order of magnitude.

The fix is the one GRAVITY Collaboration et al. (2020) themselves use: fit the frame-tie parameters jointly with the orbit, constrained by their real Gaussian priors, rather than assuming them away. Extending the model to 11 parameters ($x_0, y_0, \dot{x}_0, \dot{y}_0$ added, Gaussian-penalized against the P. M. Plewa et al. priors) and refitting the same drift-injected data recovers $\dot{\omega}$ to within 0.9% of truth in both cases—the frame parameters themselves come back consistent with their injected values to within the prior width. Propagating this properly (bootstrap over the 11-parameter fit) gives a final $\dot{\omega}$ precision of 0.0021 – 0.0022 deg yr $^{-1}$, $\sim 1.5\times$ wider than Table 3’s frame-perfect value (0.0014 deg yr $^{-1}$) but two orders of magnitude better than the naive fit’s unmitigated bias. As with the cadence design (Section 3), a naive approach fails badly and invisibly, while explicitly modeling the known physical/instrumental degeneracy—using real, published priors rather than assuming a perfect instrument—recovers the result.

4.3. Systematic Error Budget: Source Confusion/Crowding

The remaining instrumental term is source confusion: any neighboring star within roughly one diffraction-limited beam blends into the same single-mode-fiber output and biases the recovered photocenter by (to leading, instrument-agnostic order) its flux-weighted contribution. We bound this with a Monte Carlo (`confusion_error.py`): random realizations of the local field around S301 ($m_K = 19.3$), drawing neighbor counts and positions from GRAVITY Collaboration et al. (2022)’s own quoted stellar surface density (“>100 stars per square arcsecond to $K < 17$,” with the number-magnitude slope out to $K = 21$ taken from G. F. Rubilar & A. Eckart’s quoted Jaroszyński 1999 ratio, $\nu_{21} \approx 25 \times \nu_{17}$, applied to this updated normalization), weighted by a Gaussian beam-coupling kernel matched to GRAVITY Collaboration et al. (2020)’s quoted 60 mas single-UT FWHM.

Applied naively (a single-aperture imaging photocenter, with no interferometric advantage), this gives a median bias of ~ 9.7 mas per epoch—wildly larger than GRAVITY’s real, demonstrated $\sim 100 \mu\text{as}$ precision, confirming that the naive photocenter number is the wrong quantity to use directly. GRAVITY Collaboration et al. (2022) state interferometric astrometry is “much less affected (by a factor of several hundred) by confusion noise” than this naive imaging case; bounding that qualitative range as $200\text{--}500\times$, the actual GRAVITY confusion floor (90th-percentile realization) is $37\text{--}92 \mu\text{as}$, comparable to or somewhat below the $100 \mu\text{as}$ photon-noise floor already modeled in Table 3.

Unlike the frame tie (Section 4.2), confusion is not a coherent drift across the campaign’s baseline—neighboring S-cluster stars move on their own orbits, so a given blend configuration is not persistent between widely separated epochs. We therefore propagate it as additional per-epoch scatter (quadrature-combined with the existing $100 \mu\text{as}$ floor) rather than a coherent offset, and re-run the real bootstrap fit with this inflated sigma: even at the conservative end ($200\times$ suppression, 90th-percentile realization), $\dot{\omega}$ ’s bootstrap precision is unchanged to 3 significant figures ($0.0014 \text{ deg yr}^{-1}$, a $1.00\times$ ratio), and the best-fit value shifts by $< 10^{-4} \text{ deg yr}^{-1}$ —consistent with re-optimization noise, not a systematic bias. Confusion/crowding is real but, for this campaign’s interferometric technique, subdominant to the already-modeled random error.

4.4. Beyond the Error Budget: Spin (Lense-Thirring) Contamination

The three systematics above share a structure: each has a known or boundable magnitude, so each can be modeled, subtracted, or shown subdominant. Sgr A*’s spin is different in kind. For a spin aligned with S301’s orbital angular momentum, frame-dragging produces a purely apsidal shift (`orbit.lense_thirring_apsidal_rate`, validated against GRAVITY Collaboration et al.’s own quoted 0.114 deg/orbit at $\chi = 1$)—i.e., it adds directly and indistinguishably to the same $\dot{\omega}$ this entire pipeline fits for, with no analogous upper bound to inject the way GRAVITY Collaboration et al. (2022)’s extended-mass limit was used in Section 4.1.

Sgr A*’s spin is not robustly measured; real estimates vary by method. Using GRAVITY Collaboration et al. (2026)’s own fiducial benchmark ($\chi = 1$, aligned) and an independent radio/X-ray outflow-method measurement ($\chi = 0.90 \pm 0.06$; R. A. Daly & et al. 2024), the possible Lense-Thirring contamination of $\dot{\omega}$ is $4.7\text{--}9.5\sigma$ relative to this project’s random-error precision ($2.9\text{--}5.7\%$ of the Schwarzschild signal itself)—larger, in sigma terms, than any of the three bounded systematics after their respective fixes. A measured $\dot{\omega}$ from S301 alone cannot distinguish a pure point-mass GR test from GR plus aligned frame-dragging; this is exactly the degeneracy T. Piran et al. (2026)’s multi-star differential method exists to break, and exactly why this paper’s own spin-agnostic framing (Section 5, *Scope*) is a scope choice, not a claim that the reported $\dot{\omega}$ is free of this effect.

4.5. A Geometric Effect: Roemer (Light-Travel-Time) Delay

`orbit.py`’s engine explicitly excludes Roemer delay (Section 3.1): the light-travel-time from S301 to the observer varies over the orbit as the star’s line-of-sight distance changes, so an observed arrival time does not equal the dynamical emission time the model assumes. S301’s orbital diameter implies a maximum possible swing of $2a/c \approx 7.9$ days—comparable to the campaign’s own ± 20 -day periastris-dense window (Section 3); across the real campaign epochs the actual correction ranges -0.06 to $+5.75$ days (RMS 3.48 days).

We tested this directly (`roemer_delay.py`): building a noise-free synthetic dataset with the true light-time correction applied, then refitting with the unmodified (light-time-naïve) 7-parameter model. The effect on $\dot{\omega}$ is modest—a bias of $-0.0017 \text{ deg yr}^{-1}$ (0.75% of the signal, 1.3σ relative to Table 3’s random-error precision)—but the effect on the orbit’s orientation angles is not: i , Ω , and ω are biased by $+0.49^\circ$, -1.31° , and -0.70° respectively, corresponding to $14\text{--}16\sigma$ relative to their own bootstrap uncertainties. Unlike the other systematics in this budget, Roemer delay barely touches the parameter this paper is built around, while substantially distorting the recovered orbital geometry itself.

Table 4. Comprehensive error budget for $\dot{\omega}$.

Term	Type	Effect (deg yr ⁻¹ , σ)	Status
Photon noise	Random	± 0.0014 (1σ)	Baseline (Table 3)
Extended mass	Astrophysical	-0.0001 to -0.0003 (0.1 – 0.2σ)	Subdominant, bounded
Frame tie, unmitigated	Instrumental	-0.038 to -0.107 (27 – 77σ)	Dominant if ignored
Frame tie, joint fit	Instrumental	residual 0.9% ; precision 0.0021 – 0.0022	Fixed w/ real priors
Source confusion	Instrumental	$< 1\%$ precision change	Subdominant
Roemer delay, unmitigated	Geometric	-0.0017 (0.75% , 1.3σ)	Fails if ignored
Roemer delay, corrected	Geometric	residual 0 (exact)	Fixed w/ correct model
Sgr A* spin	Degeneracy	$+0.0066$ to $+0.0133$ (4.7 – 9.5σ)	Unbounded, out of scope

NOTE—Sections 4.1 (extended mass), 4.2 (frame tie), 4.3 (confusion), 4.4 (spin), and 4.5 (Roemer delay) give each term’s real citation sources and methodology. Unmitigated Roemer delay also biases the orbital orientation angles i, Ω, ω by 14 – 16σ , not shown above; this is corrected along with $\dot{\omega}$ once the model accounts for light time.

The correct fix requires no new free parameters, unlike the frame tie’s real nuisance parameters (Section 4.2): light time is a deterministic function of the orbit already being fit, so a self-consistent light-time correction inside the model (solving $t_{\text{emit}} = t_{\text{obs}} + z(t_{\text{emit}})/c$ for each trial parameter vector) is, in principle, exact. We verified this directly at the true parameters—the corrected model reproduces the noise-free data to machine precision (zero residual)—confirming the bias above is a modeling omission, not a data artifact. Robustly *recovering* that optimum via nonlinear least-squares initially failed for an unrelated reason: tracing the failure down to individual observation epochs uncovered a genuine bug in `orbit.py`’s shared Kepler-equation solver, used by every calculation in this project. Undamped Newton-Raphson from the naive starting guess $E_0 = M$ —this function’s original implementation—diverges catastrophically and permanently for a subset of ordinary (mean anomaly, eccentricity) pairs at S301’s $e = 0.9832$ (verified directly: $M = 0.2185$ rad, an unremarkable value, blows up to $E \sim 10^{13}$ and never recovers within the full 200-iteration budget). This had not been visible elsewhere in this project because the true orbital elements’ own epochs happen to avoid the divergent region, but a nonlinear optimizer exploring nearby trial parameters—exactly what fitting does—routinely does not. We replaced it with a bisection-safeguarded Newton solve (a standard globally convergent technique, since $E - e \sin E - M$ is strictly monotonic and a containing bracket $[M - e, M + e]$ is known a priori), verified against 1.4×10^6 random (mean anomaly, eccentricity) pairs up to $e = 0.9999$ (worst-case residual 2×10^{-12}), and confirmed to leave every other result in this paper unchanged to within ordinary bootstrap resampling noise (Table 3’s $\dot{\omega}$ precision shifts from 0.0013 to 0.0014 deg yr⁻¹, consistent with a small number of previously silently corrupted bootstrap resamples now computed correctly). With the underlying solver fixed, the Roemer-aware fit converges to the true parameters exactly (zero residual to machine precision)—this systematic is fully corrected, the same way the frame tie was in Section 4.2, once the model is right.

5. DISCUSSION

Comprehensive error budget. Table 4 collects every term quantified in Sections 4.1–4.5, alongside the random error from Table 3. Two things do not fit into a single “total uncertainty” number, and conflating them would misstate what this pipeline has actually shown:

First, the random and instrumental terms combine coherently once mitigated: the frame-tie joint fit’s 0.0021 – 0.0022 deg yr⁻¹ already subsumes photon noise, and adding the extended-mass and confusion contributions in quadrature moves it by $< 1\%$ (e.g. $\sqrt{0.0021^2 + 0.00013^2} = 0.002104$)—so ~ 0.0021 deg yr⁻¹ is the honest bottom-line *precision* of this pipeline, roughly 1.5 – $1.6\times$ Table 3’s frame-perfect figure, not the number itself. Roemer delay, once a real, uncorrected bias of comparable size, is now fully removed once the model accounts for light time (§4.5)—the correct treatment, once implemented, requires no ongoing quadrature term any more than the frame tie’s does once jointly fit.

Second, and most importantly, none of this bounds the spin systematic, which is 3 – $6\times$ larger than the mitigated precision and has no analogous limit to inject. This is the fundamental reason the result is described as a *spin-agnostic*

test throughout this paper, not a spin-free one: at real, plausible spin values, a measured $\dot{\omega}$ could differ from the pure-Schwarzschild prediction by more than everything else in Table 4 combined. Closing this gap is exactly T. Piran et al. (2026)’s multi-star program, not a numerical refinement of the single-star pipeline presented here.

Instrumentation: astrometry. No operating alternative beats GRAVITY+ for the astrometric measurement this paper actually proposes. The comparison is a matter of raw resolution: Roman’s 2.4 m aperture gives a diffraction limit of order 223 mas, larger than S301’s entire 83 mas orbit, so it cannot resolve the orbit’s shape at all, let alone its precession. JWST’s larger 6.5 m aperture is sharper (~ 85 mas) but is still a single telescope, not an interferometer, so its resolution remains comparable to the orbit’s own angular scale rather than the small fraction of it GRAVITY+’s ~ 130 m effective interferometric baseline achieves ($\sim 100 \mu\text{as}$, Section 3).

Instrumentation: radial velocity. No alternative helps here either, for three separate reasons. ERIS/SPIFFIER itself cannot: Section 3 shows its real, achieved S2 precision, scaled to S301’s fainter magnitude, predicts $\sim 1621 \text{ km s}^{-1}$ against a $\pm 15,000 \text{ km s}^{-1}$ signal ($\text{SNR} \approx 0.04$)—this is why RV was never proposed for the actual campaign in the first place, not a capability that was tried and abandoned. Roman fails independently of integration time: its PSF is not adaptive-optics-corrected, so source confusion in this crowded field is a structural floor that does not average down with longer exposures, compounded by its coarser spectral resolution ($R \approx 461\text{--}890$ vs. ERIS’s 5000). JWST/NIRSpec is more ambiguous—its real achieved resolution is comparable to or better than pre-launch expectations (A. J. Shajib et al. 2025), and it has no ground-based OH airglow to contend with—but real observations of this specific field report severe, Galactic-Center-specific operational problems (detector saturation on bright cluster members, cross-talk between micro-shutters in the field, and misidentification of guide stars in the crowded core; F. Yusef-Zadeh et al. 2025), and no published achieved precision exists yet for a point source as faint as S301, so we do not adopt it as a demonstrated alternative.

Instrumentation: what HARMONI on the ELT could add. HARMONI is worth a closer look than a schedule dismissal, because it is not just a bigger version of GRAVITY+—its 39 m aperture and K-band integral-field spectroscopy at up to $R \approx 17,385$ (N. Thatte & et al. 2024) address the RV problem specifically, not just astrometry. Extending this project’s own validated RV scaling (Section 3: a real achieved SINFONI/ERIS result, scaled by flux in the same background-limited regime documented for this field) to HARMONI’s much larger collecting area and finer spectral resolution predicts $\sigma_{\text{RV}} \approx 21 \text{ km s}^{-1}$ —against S301’s $\pm 15,000 \text{ km s}^{-1}$ signal, an SNR of order 10^3 , i.e., genuinely feasible RV, not a marginal improvement. This is a first-principles extrapolation to an instrument that has not yet observed anything, not a scaling of an achieved HARMONI result the way the ERIS number is, so it should be read as motivation for follow-on work, not a proposal-ready sensitivity figure. On timeline: ESO’s latest published schedule has HARMONI’s science first light in December 2030 (ESO 2026), about ten months before passage 1 (October 2031). That is a tight window for a new instrument to clear proposal review, but not obviously an impossible one if preparatory work starts now and no further delay occurs—the honest statement is that passage 1 is a stretch goal for HARMONI, while passage 2 (2040.5) is comfortably within reach, and the possibility of adding a real RV channel by then is exactly the kind of advantage worth planning a proposal around rather than writing off.

Scope. Measuring Sgr A*’s spin is explicitly outside this work, and so, for now, is real data: like this paper, T. Piran et al. (2026) is a forecast, not a completed measurement—no periapsis passage has occurred yet for either analysis to use. Spin causes a distinct signature (nodal precession from frame-dragging) not injected into this project’s synthetic campaign; T. Piran et al. (2026) present the methodology a real spin measurement would additionally require, separating that signal from Newtonian confusion via multi-star differential calibration, a substantially harder analysis than the single-star, spin-agnostic one presented here. Concretely, this study’s contribution to that eventual measurement is the part it actually completes: a validated light-curve and astrometric-fitting pipeline, a demonstrated observing cadence, and the systematic error budget in Table 4—the groundwork the harder, multi-star spin analysis would be built on top of, not a step of that analysis itself.

Design lesson. Local Fisher-information design assumes the model is well approximated by its linear expansion across the *entire* candidate range—an assumption that fails badly, invisibly to the analytic bound, for nonlinear periodic problems evaluated over multi-cycle baselines. We recommend pairing any such design with pseudo-Bayesian prior-averaging (K. Chaloner & I. Verdinelli 1995), an explicit domain-knowledge floor against non-identifiability, and validation against a real nonlinear fit rather than trust in the analytic prediction alone.

6. SUMMARY

A relativistic light-curve model reproduces S301’s published solution to $<0.1\%$ and predicts an asymmetric beaming feature, not an occultation dip, that survives visibly inside realistic noisy photometry. A feasibility calculation shows ERIS cannot detect S301’s RV at any useful precision (Section 5), which is why this campaign proposes astrometry only rather than including a channel known in advance to fail. A naive Fisher-optimal cadence catastrophically fails via nonlinear aliasing despite an excellent analytic prediction; a hybrid design (domain-knowledge floor + Fisher-optimal remainder) fixes this and beats a simple heuristic on every parameter. The resulting astrometry-only, two-passage GRAVITY+ campaign recovers the injected Schwarzschild precession rate ($0.2307 \text{ deg yr}^{-1}$) as $0.2326 \pm 0.0014 \text{ deg yr}^{-1}$, and a non-circular check confirms this geometric result independently agrees with what the 1PN formula predicts from the same fit’s own period and eccentricity (0.91σ), without either quantity referencing the other. A numerical post-Newtonian integration of S301’s real orbit, using G. F. Rubilar & A. Eckart (2001)’s methodology and GRAVITY Collaboration et al. (2022)’s current extended-mass bound, shows the leading astrophysical systematic (Newtonian confusion) biases $\dot{\omega}$ by only $0.09\text{--}0.23\sigma$ of the random measurement error even at the conservative 3σ mass bound—subdominant. The leading instrumental systematic is not: the astrometric reference-frame tie, at the real magnitude GRAVITY Collaboration et al. (2020) report as limiting their own S2 measurement, biases a naive fit’s $\dot{\omega}$ by $16\text{--}46\%$ of the entire signal ($27\text{--}77\sigma$) if left unmodeled—but fitting it jointly with real Gaussian priors, exactly as GRAVITY Collaboration et al. (2020) do, recovers $\dot{\omega}$ to within 0.9% of truth, at a final precision ($\sim 0.0021 \text{ deg yr}^{-1}$) only $\sim 1.5\times$ wider than the frame-perfect case. A Monte Carlo bound on the remaining instrumental term, source confusion, shows GRAVITY’s real, quoted interferometric advantage ($200\text{--}500\times$ less affected than single-aperture imaging) keeps it comparable to or below the photon-noise floor already modeled, changing $\dot{\omega}$ ’s precision by $<1\%$. Two further checks probe the edges of this budget: an aligned Sgr A* spin, at real published estimates ($\chi = 0.9\text{--}1$), would contaminate $\dot{\omega}$ at the $5\text{--}10\sigma$ level with no analogous bound to subtract—the genuine, unresolved reason this remains a spin-agnostic test rather than a spin-free one; and a real Roemer (light-travel-time) delay, up to several days for this orbit, would bias $\dot{\omega}$ and distort the recovered orbital orientation by up to 16σ if ignored, but is fully corrected by a self-consistent light-time model (zero residual bias)—a fix that, in the process of being made to converge robustly, surfaced and repaired a genuine divergence bug in the project’s core Kepler-equation solver, verified not to otherwise change any result in this paper beyond ordinary bootstrap noise. This is a synthetic forecast, not a completed measurement: real confirmation awaits the actual 2031.8 and 2040.5 passages. We conclude that, once observed, a two-passage GRAVITY+ campaign would be feasible with existing instrumentation and sufficient for an independent, spin-agnostic GR test near Sgr A*, ahead of the harder spin measurement S301’s discovery motivated.

This paper’s contribution is best understood as two things, not one. The first is the feasibility case and error budget themselves: a numerically verified demonstration that existing instrumentation suffices, paired with a systematic-error treatment—extended-mass confusion, the astrometric frame tie, source confusion, Roemer delay, and the spin degeneracy—that is more complete on this specific question than either real S301 paper it builds on. GRAVITY Collaboration et al. (2026) forecasts spin sensitivity without a systematic error budget; T. Piran et al. (2026) addresses the harder, related problem of how to separate spin from Newtonian confusion across multiple stars once real data exist, not the error budget for a single star’s monopole term. Neither is a gap in that work—their questions are different from this paper’s—but it means the numbers in Table 4 did not exist anywhere in the literature prior to this analysis, for this star and this measurement.

The second is methodological, and emerged rather than being planned. The same failure pattern surfaced three independent times: a Fisher-information cadence design that was excellent analytically and catastrophic in a real nonlinear fit (Section 3); an astrometric frame tie that, ignored, biased $\dot{\omega}$ by tens of percent despite the underlying data being fine (Section 4.2); and a Kepler-equation solver that diverged silently for ordinary parameter values a fitting routine would actually visit, caught only because a robust fit was demanded of it (Section 4.5). In each case the fix was not cleverness but refusal to trust an analytic or first-pass result without a real nonlinear stress test. That pattern, not any single number in Table 4, is the more transferable lesson: for this class of problem—nonlinear, periodic, evaluated over a multi-cycle baseline, at the edge of an instrument’s design regime—an untested analytic shortcut is not a conservative default. It is a specific, identifiable risk.

ACKNOWLEDGMENTS

This work uses the published orbital solution of [GRAVITY Collaboration et al. \(2026\)](#), and engages with [T. Piran et al. \(2026\)](#)’s multi-star Newtonian-confusion methodology for comparison in Section 4.1, without adopting it directly. Code, figures, and the precession animation are available at <https://github.com/jcatanza/s301-schwarzschild-precession>.

Simulation code implementation, data analysis, and manuscript drafting for this work were developed with substantial assistance from Claude Sonnet 5 (Anthropic), a large language model, under the direction, review, and final approval of the author. All scientific claims, numerical results, and citations were independently verified by the author against the underlying code output and cited primary sources before inclusion.

REFERENCES

- Chaloner, K., & Verdinelli, I. 1995, *Statistical Science*, 10, 273
- Daly, R. A., & et al. 2024, *Monthly Notices of the Royal Astronomical Society*, 527, 428.
<https://arxiv.org/abs/2310.12108>
- Davies, R. I., & et al. 2023, *Astronomy & Astrophysics*, 674, A207. <https://arxiv.org/abs/2304.02343>
- ESO. 2026, Telescope first light for ESO’s Extremely Large Telescope now planned for March 2029,,
<https://www.eso.org/public/announcements/ann25001/>
- Event Horizon Telescope Collaboration. 2022, *The Astrophysical Journal Letters*, 930, L12,
doi: [10.3847/2041-8213/ac6674](https://doi.org/10.3847/2041-8213/ac6674)
- GRAVITY+ Collaboration. 2023, arXiv e-prints.
<https://arxiv.org/abs/2301.08071>
- GRAVITY Collaboration, Abd El Dayem, K., & et al. 2026, *Nature*. <https://arxiv.org/abs/2607.12664>
- GRAVITY Collaboration, Abuter, R., Amorim, A., & et al. 2018, *Astronomy & Astrophysics*, 615, L15.
<https://arxiv.org/abs/1807.09409>
- GRAVITY Collaboration, Abuter, R., Amorim, A., & et al. 2020, *Astronomy & Astrophysics*, 636, L5.
<https://arxiv.org/abs/2004.07187>
- GRAVITY Collaboration, Abuter, R., Amorim, A., & et al. 2022, *Astronomy & Astrophysics*, 657, L12.
<https://arxiv.org/abs/2112.07478>
- GRAVITY Collaboration, Abuter, R., & et al. 2017, *Astronomy & Astrophysics*, 602, A94.
<https://arxiv.org/abs/1705.02345>
- Piran, T., Amaro-Seoane, P., Aytac, B., & et al. 2026, arXiv e-prints. <https://arxiv.org/abs/2607.24931>
- Plewa, P. M., Gillessen, S., Eisenhauer, F., & et al. 2015, *Monthly Notices of the Royal Astronomical Society*, 453, 3234. <https://arxiv.org/abs/1509.01941>
- Rubilar, G. F., & Eckart, A. 2001, *Astronomy & Astrophysics*, 374, 95, doi: [10.1051/0004-6361:20010640](https://doi.org/10.1051/0004-6361:20010640)
- Shajib, A. J., Treu, T., Melo, A., & et al. 2025, *Astronomy & Astrophysics*, 702, L12.
<https://arxiv.org/abs/2507.03746>
- Thatte, N., & et al. 2024, in *Proc. SPIE*, Vol. 13096, 1309614, doi: [10.1117/12.3018520](https://doi.org/10.1117/12.3018520)
- Yusef-Zadeh, F., Bushouse, H., Arendt, R. G., & et al. 2025, *The Astrophysical Journal Letters*.
<https://arxiv.org/abs/2501.04096>